

SANJAY VENKATESAN

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PROFESSIONAL SUMMARY

Results-driven professional combining a B.Tech in Mechanical Engineering with expertise in medical insurance operations, robotics instruction, and full-stack web development. Experienced in processing insurance claims, coordinating with healthcare providers, and delivering STEM education through hands-on robotics projects. Proficient in CAD/CAM, cloud technologies, and modern programming frameworks. Committed to precision, continuous learning, and creating impactful solutions across technical and operational domains.

SKILLS

Technical: CAD/CAM · ROBCAD · Arduino · Full-Stack Dev (Python, Java) · HTML/CSS/JS · Bootstrap · Figma · Adobe Illustrator · Google Cloud AI

Insurance Operations: Claims Processing · Policy Administration · Provider Coordination · Compliance · Data Verification

Soft Skills: Communication · Time Management · Decision Making · Active Listening · Classroom Management · Creative Thinking

PROFESSIONAL EXPERIENCE

Robotics Instructor — *Lurn X, Chittoor, India*

Feb 2024 – Present

- Designed and delivered a comprehensive robotics curriculum covering Arduino programming, sensor integration, and autonomous systems for students across multiple grade levels.
- Guided students through hands-on projects including robot design, fabrication, and testing — fostering practical problem-solving and engineering thinking.
- Developed STEM-aligned lesson plans incorporating real-world robotics applications and project-based learning modules.
- Established productive relationships with students, parents, teachers, and administrators to support holistic student development.
- Counselled students facing educational and personal challenges, creating an inclusive and encouraging learning environment.
- Evaluated student progress through structured assessments and delivered constructive feedback to drive continuous improvement.

Technical Trainee — *Claims & Data Operations MediAssist Healthcare TPA Pvt. Ltd., India*

Sep 2022 – Dec 2022

- Systems & Data Management:** Managed high-volume claim data within complex Claims Management Systems (CMS), ensuring 100% data integrity, accuracy, and adherence to company protocols.
- Analytical Reporting:** Developed and automated daily/weekly MIS reports using internal tools to track KPIs, including claim processing turnaround times (TAT) and error rates, directly supporting operational decision-making.
- Business Logic & Compliance:** Interpreted and applied complex policy logic and regulatory frameworks (IRDAI) to automated workflows, ensuring technical compliance and seamless system integration between healthcare providers and internal databases.
- Troubleshooting & Resolution:** Acted as a technical point of contact for external stakeholders to resolve billing discrepancies, validate diagnostic data, and troubleshoot documentation issues within the digital claims portal.
- Quality Assurance:** Conducted rigorous data validation of electronic health records (EHRs) and billing logs to detect anomalies, ensuring that all processed data met pre-defined system validation rules and security standards.
- Process Optimization:** Collaborated with cross-functional teams to identify bottlenecks in the claims lifecycle, contributing to improved workflow efficiency and system uptime.

EDUCATION

B. Tech in Mechanical Engineering	2022
<i>Coursework: Engineering Mechanics, Thermodynamics, Fluid Mechanics, Heat & Mass Transfer, Manufacturing Technology, CAD/CAM, Strength of Materials.</i>	
<i>Dissertation: "Design and Fabrication of HSS and Carbide Boring Drilling Bit" — CAD optimization, performance analysis, wear resistance & material comparison.</i>	
Intermediate — MPC	2017
Secondary School Certificate (SSC)	2015

CERTIFICATIONS

Certified Full Stack Developer — Python & Java: Comprehensive training in front-end and back-end development using Python (Django/Flask) and Java (Spring Boot), RESTful APIs, databases, and deployment practices.

Google Cloud AI Essentials: Google-issued certification covering foundational AI/ML concepts, Vertex AI, BigQuery ML, and responsible AI practices on Google Cloud Platform.

CAD, CAM & ROBCAD Certification: Andhra Pradesh State Skill Development Corporation (APSSDC) — certified in computer-aided design, manufacturing, and robotic simulation tools.

KEY PROJECTS

Automated Grass Cutter Robot: Engineered a fully autonomous grass-cutting robot using Arduino UNO, ultrasonic sensors, and motor control logic. Integrated self-turning wheels, obstacle detection, and a vacuum mechanism for efficient grass collection without human intervention.

Design & Fabrication of HSS and Carbide Boring Drilling Bit: Final-year engineering project comparing boring tools made from HSS and carbide. Conducted CAD modelling, fabrication, and testing to evaluate durability, wear resistance, and cutting efficiency.

Personal Portfolio Website: Responsive portfolio built with HTML, CSS, Bootstrap & JavaScript; UI/UX designed in Figma with custom scroll animations and mobile optimization.

Product Landing Page: Visually engaging landing page featuring JavaScript animations and custom vector illustrations made in Adobe Illustrator.

Online Education Dashboard: Accessible, WCAG-compliant dashboard built with semantic HTML and modular CSS, ensuring inclusive user experience.

Restaurant Website: Multi-page site with menu browsing, online reservation forms, gallery, and contact sections — complete digital presence for a food-service brand.

LANGUAGES

English · Hindi · Tamil · Telugu

HOBBIES & INTERESTS

Cricket · Badminton · Reading Books